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A STUDY ON RESEARCH PAPERS

Flight Delay Prediction using Machine Learning

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Introduction

The global aviation industry serves as a cornerstone of modern transportation and economic connectivity, yet it remains significantly vulnerable to the persistent challenge of flight delays. These disruptions are not merely an inconvenience for passengers; they represent a massive financial burden, costing airlines and airports billions of dollars annually in operational inefficiencies, increased fuel consumption, and labor costs. Predicting these delays is a highly complex task because they stem from a "butterfly effect" where a minor issue such as a late arriving crew or a localized thunderstorm can ripple through an entire global flight network.

Traditional forecasting methods, which often relied on linear statistical models and isolated schedules, have proven insufficient for capturing the non-linear and spatial-temporal dependencies of air traffic. This has led to a paradigm shift toward **Machine Learning (ML)** and **Deep Learning (DL)**, which can ingest "Aviation Big Data" to identify hidden patterns. Recent advancements have expanded the scope of data beyond simple schedules to include high-frequency **ADS- B (Automatic Dependent Surveillance-Broadcast)** messages, real-time meteorological conditions, and even textual aeronautical reports like **METAR** and **NOTAMs**.

This literature review analyzes ten seminal research papers published between 2017 and 2026. These studies represent the cutting edge of aviation analytics, exploring diverse methodologies ranging from **Random Forest** and **XGBoost** to advanced **Graph Machine Learning** and **Large Language Models (LLMs)**. By examining these works, we can understand the evolution of predictive modeling from localized studies of single airlines to sophisticated, multimodal frameworks designed to navigate the complexities of modern, interconnected air traffic management (ATM) systems.

Research Paper Summaries

Article 1

Citation: T. Shruthi, D. Rajitha, and A. Shruthi, "Flight Delay Prediction Based on Aviation Big Data and Machine Learning," International Journal of Drug and Medical Device Research, vol. 5, no. 2, 2025.

This research addresses the limitations of traditional flight delay models that often focus on isolated routes or single airports. The authors developed a comprehensive predictive framework that integrates high-frequency Automatic Dependent Surveillance-Broadcast (ADS-B) messages with external environmental factors. The study emphasizes the "network effect" of aviation, where a delay in one geographical sector ripples through the system. By combining real-time flight tracking data with historical schedules and meteorological data, the study explores both binary classification (delay vs. no delay) and regression (predicting the specific duration of the delay). A significant finding was the comparative failure of LSTM models in this specific architectural setup due to overfitting, leading to the proposal of an optimized ensemble approach.

Dataset Used: A multi-source "Aviation Big Data" repository. This includes real-time ADS-B telemetry (position, altitude, velocity), historical flight schedules, airport metadata, and detailed weather records (visibility, precipitation, wind speed, and temperature). The data spans multiple major air traffic corridors to ensure a diverse representation of flight conditions.

RL Environment Used: Not applicable. This study utilizes a supervised learning framework focusing on classification and regression tasks rather than reinforcement learning.

Algorithm / Methodology Used: The methodology follows a robust data science pipeline:

1. **Data Integration:** Fusing ADS-B messages with weather and schedule data.
2. **Preprocessing:** Feature scaling, handling missing telemetry data, and categorical encoding for airport codes.
3. **Model Selection:** Comparison of K-Nearest Neighbors (KNN), Long Short-Term Memory (LSTM) networks, and Random Forest.
4. **Validation:** 80/20 train-test split with cross-validation to ensure model stability across different time-slices of flight data.

Performance Metrics / Results: * Binary Classification Accuracy: 90.2% achieved by the optimized Random Forest model.

Limitations of This Work: The primary limitation is the sensitivity of the model to ADS-B "dark spots" or signal loss in certain geographical areas. Additionally, while the model accounts for weather, it does not currently integrate non-physical factors such as Air Traffic Control (ATC) staffing levels, technical maintenance logs of specific airframes, or unexpected security events, which can be major contributors to irregular flight operations.

Article 2

Citation: G.W.I. Dilshani and M. Tissera, "Airline Delay Prediction: A Comparative Study using Machine Learning and Deep Learning Models," University of Kelaniya, 2024.

This paper focuses on bridging the gap between theoretical ML models and practical airline operational tools. The researchers conducted an exhaustive comparative study of five different algorithmic families to determine which is most resilient to the noise found in commercial airline data. A major portion of the work is dedicated to the "Hyperparameter Optimization" phase, utilizing grid search and cross-validation to squeeze maximum performance out of the models. The study concludes that deep architectural approaches, specifically Multilayer Perceptrons, can capture the subtle, non-linear relationships between transit airport congestion and final arrival delays more effectively than traditional linear classifiers, provided they are tuned correctly.

Dataset Used: A large-scale dataset including flight timings, travel duration, transit airport details, and historical delay patterns.

RL Environment Used: Not applicable. Supervised classification and regression.

Algorithm / Methodology Used: Random Forest, Logistic Regression, Gradient Boosted Trees, Support Vector Machines (SVM), and Multilayer Perceptron (MLP); utilized extensive hyperparameter tuning.

Performance Metrics / Results: The Multilayer Perceptron (MLP) model achieved the highest performance with a prediction accuracy of 98.83%.

Limitations of This Work: The study primarily focuses on algorithmic tuning and does not deeply integrate real-time external disruptions like sudden airspace closures or complex gate-assignment conflicts.

Article 3

Citation: M. Mamdouh, M. Ezzat, and H. A. Hefny, "A novel intelligent approach for flight delay prediction," Journal of Big Data, vol. 10, no. 179, 2023.

The authors propose the "Flight Delay Path Previous-based Machine Learning" (FDPP-ML) algorithm to solve the data accessibility problem in aviation. Recognizing that private airline data and high-resolution weather feeds are often proprietary or expensive, this study focuses on achieving high-precision regression using only basic, publicly available flight schedule features. The FDPP-ML algorithm treats the flight network as a sequential path, using the delay status of previous "legs" in an aircraft's journey to predict future performance. This "path-based" logic allows the model to maintain high accuracy even when environmental data is sparse, making it a highly portable solution for various global flight networks.

Dataset Used: Historical flight and weather data (sourced from Kaggle and FAA Core 30 records), focusing on spatial- temporal correlations within the flight network.

RL Environment Used: Not applicable.

Algorithm / Methodology Used: Flight Delay Path Previous-based Machine Learning (FDPP-ML) algorithm; supervised regression models.

Performance Metrics / Results: Improved individual delay minute prediction with high regression accuracy compared to baseline ML models.

Limitations of This Work: By relying on basic schedule features, the model may lose precision during rare, high-impact weather events that deviate from historical patterns.

Article 4

Citation: S. P. Pophale, P. R. Patil, A. D. Potgantwar, and P. R. Bhaladhare, "Flight Delay Analysis and Prediction Using Machine Learning Algorithms," Mathematical Statistician and Engineering Applications, vol. 71, no. 4, 2022.

This study approaches flight delays from two distinct angles: departure management and arrival forecasting. The authors argue that a single model is insufficient to capture the different dynamics of a flight's lifecycle. They developed a dual- model framework where the first model uses regression to predict the exact number of minutes a flight will be delayed at departure, while the second model uses classification to categorize the impact of that departure delay on the final arrival (e.g., "on time," "minor delay," or "major delay"). This allows airline agencies to not only know if a flight will be late but also to understand the severity of the operational disruption for ground crews.

Dataset Used: Historical flight performance records and environmental parameters including season and flight distance.

RL Environment Used: Not applicable.

Algorithm / Methodology Used: Linear Regression for duration prediction and Random Forest for arrival delay classification.

Performance Metrics / Results: Identified strong correlations between departure lag and final arrival delay; provided high classification precision for arrival categories.

Limitations of This Work: The model is built on static historical data and lacks a dynamic feedback loop for mid-flight updates.

Article 5

Citation:M. H. A. A. Chahine, R. Hasan, and W. A. Iddin, "Flight Delays Prediction by using Machine Learning," Global Scientific Journals, vol. 11, no. 6, 2023.

This research investigates the commercial application of ML to improve the "Passenger Experience Index." By analyzing data from a major US carrier, the authors demonstrate how predictive analytics can be used to provide passengers with more accurate departure expectations, thereby reducing airport terminal congestion. The study evaluates the trade-off between model complexity and interpretability, concluding that Ensemble methods like Random Forest provide the best balance for commercial airline deployments. The authors also explore how "Feature Importance" analysis can help airline managers identify the specific months or times of day that contribute most to their delay statistics.

Dataset Used: Historical flight schedules and weather data from a major US carrier.

RL Environment Used: Not applicable.

Algorithm / Methodology Used: Random Forest, Support Vector Machine (SVM), and Logistic Regression.

Performance Metrics / Results: Demonstrated that Random Forest is highly effective for binary delay status prediction.

Limitations of This Work: The dataset is limited to a single carrier, which may not account for inter-airline delay propagation or global network constraints.

Article 6

Citation: H. Khaksar and A. Sheikholeslami, "Airline delay prediction by machine learning algorithms," Scientia Iranica, 2019.

This study is a landmark comparative analysis of how different national infrastructures impact flight delays. The researchers applied the same ML framework to two vastly different environments: the United States and Iran. The goal was to determine if delay drivers are universal or region-specific. The findings were groundbreaking: while U.S. delays are primarily driven by "Visibility" and "Air Traffic Volume," delays in the Iranian network are dominated by "Fleet Age" and "Aircraft Type." This highlights that a "one-size-fits-all" ML model cannot be used globally; instead, models must be customized to reflect the specific maintenance and infrastructural realities of the region.

Dataset Used: U.S. flight datasets and a large Iranian airline network dataset.

RL Environment Used: Not applicable.

Algorithm / Methodology Used: Bayesian modeling, Decision Trees, Cluster Classification, and Random Forest.

Performance Metrics / Results: Accuracy varied by region; identified that "visibility" is the top factor in the US, while "fleet age" is the top factor in Iran.

Limitations of This Work: The hybrid models require high computational power and localized parameter tuning for every different airport network.

Article 7

Citation: M Dai, "A hybrid machine learning-based model for predicting flight delay through aviation big data," Scientific Reports (Nature), 2024.

This study introduces a sophisticated hybrid framework designed to handle massive volumes of information characteristic of aviation big data. The author argues that traditional single-model approaches suffer from "information overload" and reduced processing speeds. To solve this, the research implements a multi-stage pipeline: first, it uses ANOVA combined with Forward Sequential Feature Selection (FSFS) to strip away irrelevant data and identify the most influential delay indicators. Second, it employs the DBSCAN clustering algorithm to group similar flight patterns together, effectively decomposing a massive global problem into smaller, highly specialized sub-problems. Each cluster is then assigned a dedicated Random Forest model that has been mathematically optimized using the Coyote Optimization Algorithm (COA). This "divide and conquer" strategy not only sharpens prediction accuracy but also drastically reduces the computational resources required for real-time monitoring.

Dataset Used: Large-scale aviation big data featuring multiple flight indicators.

RL Environment Used: Not applicable.

Algorithm / Methodology Used: DBSCAN clustering, ANOVA, Forward Sequential Feature Selection (FSFS), and Coyote Optimization Algorithm (COA) for weighing Random Forest trees.

Performance Metrics / Results: Increased accuracy by 2.49% and improved processing speed by 39.17% over standard models.

Limitations of This Work: The complexity of the optimization algorithm (COA) makes it difficult to implement in simple, low-resource airline systems.

Article 8

Citation: J. L. Franco et al., "Graph machine learning for flight delay prediction due to holding maneuver," arXiv:2502.04233, 2025.

This research tackles one of the most specific and costly causes of air traffic disruption: the "holding maneuver," where aircraft are forced to circle in designated airspace due to congestion. The authors argue that traditional tabular datasets (rows and columns) fail to capture the "networked" nature of airspace, where a bottleneck at one airport physically blocks the path of incoming flights. To address this, the study models the aviation system as a directed multi-graph, where airports are nodes and flight paths are edges. By using Graph Machine Learning (Graph ML) and calculating "eigenvector centrality" (a measure of how important a specific node is to the whole network), the model can predict the holding delays based on the state of neighboring airspace sectors. This spatial-aware approach allows air traffic controllers to anticipate congestion ripples before they manifest as actual delays.

Dataset Used: Multi-graph datasets where flights are edges and airports are nodes.

RL Environment Used: Not applicable.

Algorithm / Methodology Used: Graph Machine Learning (Graph ML), CatBoost, and network centrality features (eigenvector centrality).

Performance Metrics / Results: Outperformed traditional tabular models by capturing the dependencies between neighboring airspace sectors.

Limitations of This Work: Requires a graph-structured database which is more complex to maintain than standard SQL tables.

Article 9

Citation: L. Carvalho et al., "A Review on Flight Delay Prediction," arXiv:1703.06118, 2021.

This paper provides a high-level systematic mapping and taxonomy of the entire flight delay prediction landscape. Rather than proposing a single model, the authors synthesized over two decades of research to identify the shift from classical statistical regression toward modern big data analytics. The review categorizes research into three primary dimensions: Scope (individual flights vs. airport networks), Data (weather, schedules, or air traffic flow), and Computational Methods (traditional ML vs. Deep Learning). A major focus of the summary is the "data deluge" the transition from small, sampled datasets to the massive, real-time streams (like ADS-B) used today. This article serves as a foundational guide for researchers, identifying that while the industry has mastered "binary" prediction (delay vs. no delay), the current frontier lies in multi-modal integration and the prediction of specific delay durations in minutes.

Dataset Used: Metadata from over 100 research papers published between 2000 and 2020.

RL Environment Used: Not applicable.

Algorithm / Methodology Used: Taxonomy development and trend analysis.

Performance Metrics / Results: Identified a clear industry shift from statistical regression to deep learning and big data analytics.

Limitations of This Work: As a review, it does not propose a new predictive model but rather synthesizes existing ones.

Article 10

Citation: T. Phisannupawong et al., "LLM4Delay: Flight Delay Prediction via Cross-Modality Adaptation," arXiv:2510.23636, 2026.

This cutting-edge paper introduces "LLM4Delay," the first framework to treat flight delay prediction as a multimodal language problem. The authors recognize that many delay factors are hidden in "Unstructured Text," such as METAR weather reports and NOTAM (Notice to Airmen) alerts. By using Large Language Models (LLMs) and a technique called "Instance-level Projection," the framework maps numerical aircraft trajectory data (from sensors) into the same mathematical space as textual reports. This allows the model to "read" a weather warning and "see" the aircraft's path simultaneously, leading to a much deeper contextual understanding of potential disruptions than any previous numerical only model.

Dataset Used: Textual aeronautical info (METAR, NOTAMs) combined with aircraft trajectory data.

RL Environment Used: Not applicable.

Algorithm / Methodology Used: LLM-based framework with cross-modality adaptation and instance-level projection.

Performance Metrics / Results: Demonstrated superior contextual understanding compared to numerical-only ATM frameworks.

Limitations of This Work: The model is computationally expensive due to the nature of LLMs and requires high-end GPU resources for real-time inference.

References

1. T. Shruthi, D. Rajitha, and A. Shruthi, "Flight Delay Prediction Based on Aviation Big Data and Machine Learning," Int. J. Drug Med. Device Res., vol. 5, no. 2, 2025.

2. G.W.I. Dilshani and M. Tissera, "Airline Delay Prediction: A Comparative Study using Machine Learning and Deep Learning Models," University of Kelaniya, 2024.

3. M. Mamdouh, M. Ezzat, and H. A. Hefny, "A novel intelligent approach for flight delay prediction," J. Big Data, vol. 10, no. 179, 2023.

4. S. P. Pophale, P. R. Patil, A. D. Potgantwar, and P. R. Bhaladhare, "Flight Delay Analysis and Prediction Using Machine Learning Algorithms," Math. Stat. Eng. Appl., vol. 71, no. 4, 2022.

5. M. H. A. A. Chahine, R. Hasan, and W. A. Iddin, "Flight Delays Prediction by using Machine Learning," Glob. Sci. J., vol. 11, no. 6, 2023.

6. H. Khaksar and A. Sheikholeslami, "Airline delay prediction by machine learning algorithms," Scientia Iranica, 2019.

7. M. Dai, "A hybrid machine learning-based model for predicting flight delay through aviation big data," Sci. Rep., 2024.

8. J. L. Franco, M. V. M. Neto, F. A. N. Verri, and D. R. Amancio, "Graph machine learning for flight delay prediction due to holding maneuver," arXiv:2502.04233, 2025.

9. L. Carvalho, A. Sternberg, et al., "A Review on Flight Delay Prediction," arXiv:1703.06118, 2021.

10. T. Phisannupawong, J. J. Damanik, and H. L. Choi, "LLM4Delay: Flight Delay Prediction via Cross-Modality Adaptation of Large Language Models and Aircraft Trajectory Representation," arXiv:2510.23636, 2026.

Research Gap

Across the reviewed studies, flight delay prediction models have achieved high accuracy, often exceeding 90%, by utilizing ensemble and deep learning techniques. However, several critical research gaps remain:

1. **Regional and Infrastructural Bias:** Most models are trained on U.S. or European datasets where weather is the primary delay driver. These models often fail in regions like the Middle East or Asia, where fleet age, maintenance infrastructure, and geopolitical constraints play a more significant role.

2. **Static vs. Real-Time Data:** There is a heavy reliance on "clean" historical datasets. Many systems lack the ability to process live, noisy streaming data from ADS-B or IoT sensors, making them less effective for mid-flight or "on-the- fly" operational adjustments.

3. **Data Silos and Modality Gaps:** While recent papers have introduced Graph ML and LLMs, most models still operate in silos—focusing either on numerical trajectories or tabular schedules. There is a lack of integrated frameworks that simultaneously process text-based aeronautical notices (NOTAMs) and spatial air traffic congestion.

4. **Explainability and Trust:** High-performing models like LSTMs and MLPs often act as "black boxes." Without Explainable AI (XAI) to justify why a delay is predicted, air traffic controllers and airline managers are less likely to adopt these tools in high-stakes safety environments.

5. **Operational Constraints:** Many studies overlook non-physical delay factors such as crew duty hour limits, gate- assignment conflicts, and complex "reactionary" delays that ripple through a multi-airline network. Furthermore, advanced models are often too computationally heavy for deployment in airports with limited IT infrastructure.